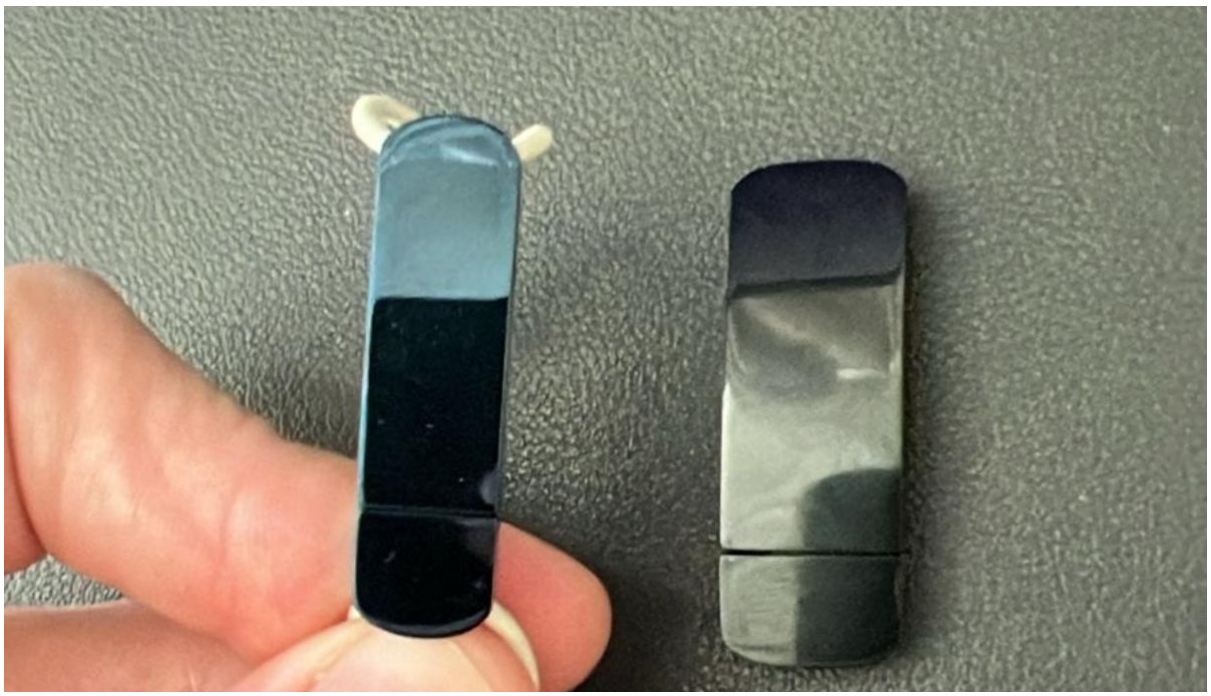
Final Product Overview:

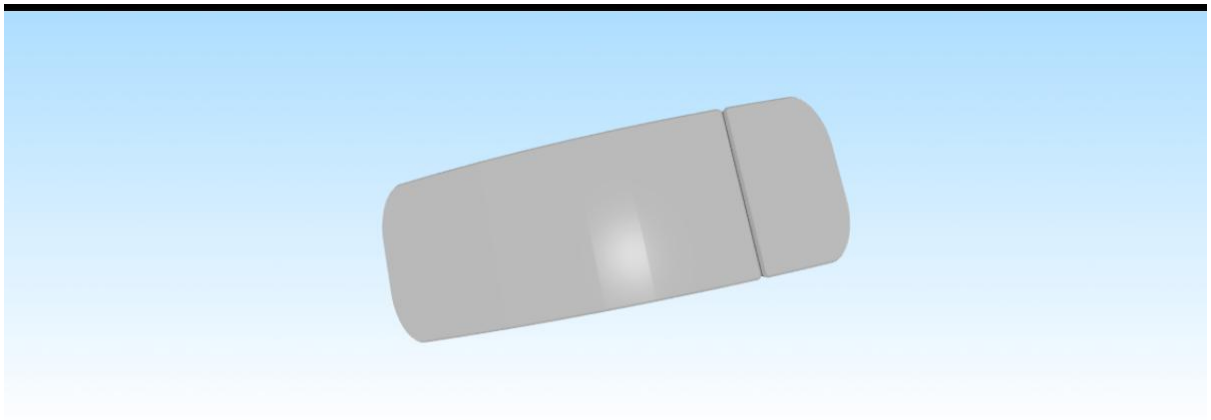
The product is a compact, battery-powered tasbeeh counter designed for long battery life, tactile feedback via vibration, and minimal visual indication via LEDs.

Operation is centered around a single button with multiple press durations controlling count, cycle selection, and reset.

The device must operate reliably within strict power, size, and usability constraints, without MCU resets, brownouts, or unintended behavior.

The below Picture shows approximately final product view.





Titanium material should be use for enclosure.

Functionality of Device:

- The device operates as a single-button electronic tasbeeh counter with visual and haptic feedback.
- The device supports four selectable counting cycles: 33, 99, 100, and 1000.
- The device increments the count by one on each short button press.
- Each valid count increment is acknowledged by a brief illumination of the orange LED.
- The device provides vibration feedback for 40ms when a counting cycle is completed.
- Upon reaching the selected cycle count:
    - A vibration event is triggered
    - The count resets automatically
    - The device immediately continues counting for the next cycle
- The device allows entry into cycle selection mode by holding the button for approximately 3 seconds.
- In cycle selection mode:
    - Short button presses step through the available cycle options
    - The currently selected cycle is indicated using the four white LEDs, displayed from left to right.
    - Other Corresponding LEDs will be OFF.

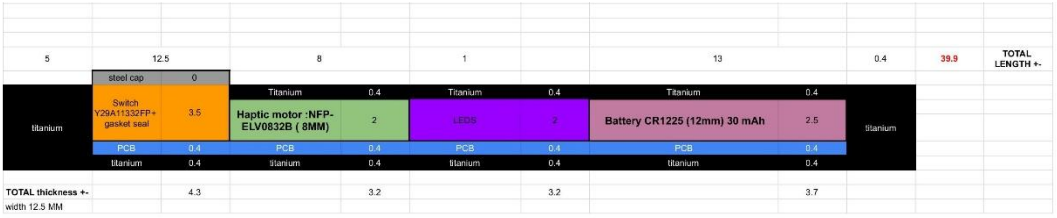
Below Pattern is priority! If it consumes more power, we shift to scheme two as describe above! But if power is not consuming much, then we will follow below pattern!

- Mode 1: LED One on!
- Mode 2: LED One and Two on!
- Mode 3: LED One Two and Three on!
- Mode 4 light One to Four on!

- If no button activity is detected for approximately 5 seconds while in cycle selection mode:
    - The selected cycle is saved
    - The count is reset
    - A distinct vibration for 80ms confirms the cycle change
- A full reset of the device (cycle and count) is performed by holding the button for approximately 6 seconds.
- The device uses four white LEDs exclusively for mode representation.
- The device uses one orange LED exclusively for:
    - Count increment indication
    - Low-battery warning

- The device monitors battery voltage and detects a low-battery condition at approximately 10% remaining capacity.
- When a low-battery condition is present and the button is pressed:
    - The orange LED illuminates to indicate low battery
    - The LED remains ON for a maximum of 10 seconds
    - The LED turns OFF automatically after the timeout
- Low-battery indication does not block or limit counting, vibration, or normal operation.
- The device provides non-blocking haptic feedback, ensuring counting remains responsive during vibration.
- The device automatically disables LEDs and the vibration motor when idle to minimize power consumption.
- The device enters a low-power sleep mode during inactivity and wakes reliably on button press.
- The device operates reliably across the full battery voltage range without unintended resets or brownouts.

Product Assembly Overview:



Main Parts to be used:

Part	Description
LCM0720A3134F	ERM vibration motor
ATTINY1616-MNR	MCU (not final)
A-SP136W1D-F02-3T x4	White LED
SML-P11DTT86 x1	Orange LED
TPS7A0218DBVR	LDO 1.8V 200mA Output (not final)
BSS138-7-F	n-Channel MOSFET
Battery Holder	To be decided according to PCB width

Battery Life Calculation:

Power consumption has been evaluated using a representative worst-case use scenario (Mode-4 completion with 1000 counts). Total energy usage per full cycle is approximately 0.043 mAh, allowing more than 1100 complete cycles on a CR1225 coin cell. The device spends most of its time in deep sleep, with short, optimized wake-ups and minimal motor activation, ensuring long battery life.